

flyPad descent calculator

The A380 OIS IN-FLT PERF DES module on the flyPad of both aircraft, computed with the FMS performance model

Branch	pr/24-efb-descent-calculator
Aircraft	A380X and A32NX (flyPad, A380 FMC, A320 MCDU)
Commits	469b9a0c8 feat(efb): descent calculator in place of the top of descent calculator
Base	pr/23-efb-landing-calculator (FMS data answers, performance store and texts)
Status	CI checks pass; merged into develop 99f00b531. Checked in the flyPad harness; in-sim test to do.

Purpose

The flyPad Top of Descent tab only applied a descent angle to a ground speed. The real A380 computes descents in the OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), with the aircraft performance, speeds and conditions.

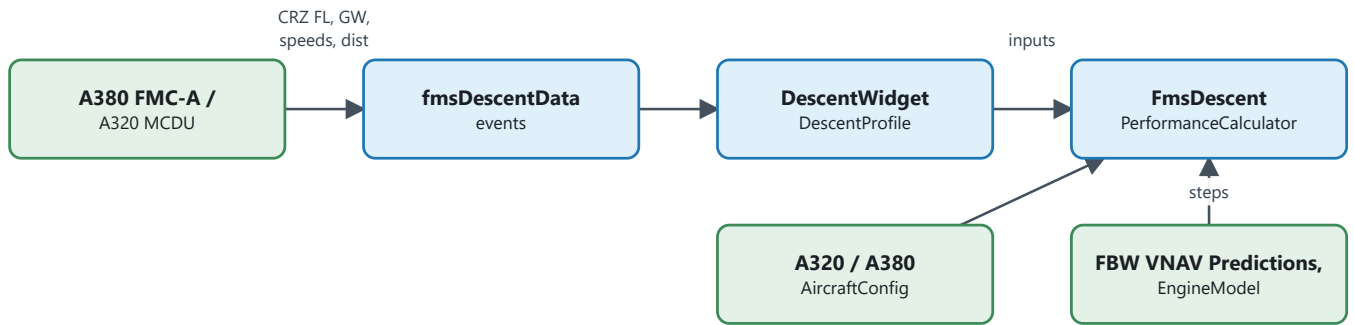
Specification (A380 FCOM PER-IFT-DES, PER-IFT-DES-STD; A320 FCOM PER-DES-STD)

Part	Content
Descent type	ECON (FMS managed Mach / CAS), STANDARD (A380 M.85 / 300 / 250 kt below FL100, A320 M.78 / 300 / 250), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes)
Descent	initial altitude (A/C), target altitude, speed brakes
Speed schedule	MACH, SPD, SPD LIM, LIM ALT; empty = the standard schedule
Conditions	gross weight, ISA deviation (A/C from the SAT), wind component, anti-ice (off / engine / total), fuel factor
FMS import	cruise level, destination elevation + 1500 ft, gross weight, speed limit, ECON speeds, distance to the destination
RESULTS	time, distance, fuel, mean V/S and FPA, 3 x altitude rule; crossover and deceleration
Profile	chart (T/D, CROSSOVER, DECEL, SPD LIM, the aircraft) or table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA)
DESCENT CHECK	distance to the target vs the profile: start in X NM (Y min), on profile, or late with the V/S and FPA needed

Model

- The FBW VNAV prediction steps (`Predictions.altitudeStep`, `verticalSpeedStep`, `speedChangeStep`, idle N1 of the engine model) with the A320 / A380 aircraft config, integrated in 1000 ft steps, so the flyPad agrees with the FMS descent predictions.
- Deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, ending at the limit altitude.
- GIVEN V/S steeper than the idle descent: idle there, with a warning (the pressure altitude below which it applies).
- Anti-ice: the corrections of the A320 FCOM descent table, marked as estimates (amber).
- Check (A320, M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg (the FCOM table is for the CFM56 A320, the model is the A320neo).

 Component added or changed by the feature  MSFS / simulator  Aircraft system (systems host, FMS, FWS)  Display / page
 Data, LVar, storage  Crew, flypad, external program



Architecture

Interfaces

Name	Type	Meaning
fms_descent_data_request	event bus	the flyPad asks the FMS for its descent data (request id)
fms_descent_data	event bus	cruise level, destination and elevation, distance, gross weight, cost index, ECON speeds, speed limit

Tests

3 unit tests in `FmsDescentPerformanceCalculator.spec.ts`: close to the A320 FCOM table; wind, temperature and anti-ice; EMERGENCY and the given V/S check.

Limitations

- The deceleration to 250 kt takes about 5000 ft, as in the FBW FMS model (a -1 degree idle path).
- No bleed in the engine model: the anti-ice effect comes from the A320 FCOM table on both aircraft.
- The old TODCalculator components are removed; its store slice is left in place, unused.

PR description (from the PR draft)

From `pr-drafts/24-efb-descent-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The Top of Descent tab of the flyPad performance page becomes a descent calculator, after the A380 OIS IN-FLT PERF DES module (A380 FCOM PER-IFT-DES), on both aircraft.

- **Descent types:** ECON (the managed descent Mach and CAS of the FMS), STANDARD (the FCOM speed schedule: A380 M.85 / 300 kt / 250 kt below FL100, PER-IFT-DES-STD; A320 M.78 / 300 kt / 250 kt, PER-DES-STD), GIVEN V/S, EMERGENCY (MMO / VMO with the speed brakes).
- **Inputs:** initial altitude (A/C: the current one) and target altitude; the speed schedule (MACH, SPD, SPD LIM, LIM ALT; empty = the standard one); gross weight, ISA deviation (A/C: from the current SAT), wind component, anti-ice (off / engine / total), fuel factor, speed brakes. **Fill data from FMS:** cruise level (or the current altitude when below), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
- **RESULTS:** time, distance, fuel, mean V/S and flight path angle, and the 3 x altitude rule for comparison; the crossover altitude and the deceleration to the speed limit.
- **Profile:** a chart of altitude against the distance to the target (T/D, crossover, DECEL, SPD LIM) or a table every 5000 ft (time, distance, fuel, CAS, Mach, TAS, V/S, FPA), like the FCOM results table.
- **DESCENT CHECK:** with the distance to the target, the aircraft altitude and ground speed: start the descent in X NM (Y min), on profile, or N NM late with the V/S and FPA needed; the aircraft is drawn on the profile.
- **Model:** the FBW VNAV prediction steps (`Predictions.altitudeStep`, `verticalSpeedStep`, `speedChangeStep`, with the idle N1 of the engine model) and the aircraft config of each aircraft, integrated in 1000 ft steps; the deceleration to the speed limit on the -1 degree idle path of the FMS descent strategy, so the flyPad agrees with the FMS predictions. GIVEN V/S steeper than idle: idle there, with a warning. Anti-ice: the corrections of the A320 FCOM descent table, shown as estimates (amber).
- **FMS import:** the A380 FMC-A and the A320 MCDU answer the flyPad (shared `fmsDescentData` events).
- The old `TODCalculator` components are removed (its `todCalculator` store slice is left in place; nothing reads it any more).
- 3 unit tests (close to the A320 FCOM descent table; wind, temperature and anti-ice; EMERGENCY and the given V/S check).

Check against the A320 FCOM descent table (M.78 / 300 / 250, 65 t, FL390 to 1500 ft): FCOM 106 NM / 17.4 min / 165 kg, calculator 114 NM / 18.8 min / 146 kg. The FCOM table is for the CFM56 A320; the FBW A320neo model descends about 8 % longer.

Cockpit API Changes

None (two new event bus topics, `fms_descent_data_request` and `fms_descent_data`).

Screenshots (if necessary)

TO ADD: the A380 screen with the profile chart and the table, the A320 screen, the DESCENT CHECK late and early.

References

- A380 FCOM PER-IFT-DES (IN-FLT PERF DES module, descent types, speed profile, conditions), PER-IFT-DES-STD.
- A320 FCOM PER-DES-STD (standard descent table and its anti-ice corrections).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad > Performance > Top of Descent, in cruise with a flight plan: Fill data from FMS: the cruise level, target, weight, speed limit and distance are filled. CALCULATE: RESULTS and the profile.
2. ECON: the speeds of the FMS; compare the T/D with the one of the ND.
3. GIVEN V/S 3000 ft/min, EMERGENCY: shorter descents; a V/S steeper than idle gives the warning.
4. DESCENT CHECK: start in X NM before the T/D; late after it, with the V/S and FPA needed.
5. `npx vitest run fbw-a32nx/src/systems/fmgc/src/performance/FmsDescentPerformanceCalculator.spec.ts` (3 tests).